

Notes: Giroud (QJE, 2013)

Proximity and Investment: Evidence from Plant-Level Data

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1 Big picture

- **Question.** Does proximity between headquarters and plants matter for plant-level investment and productivity inside multi-plant firms?
- **Setting.** The paper studies U.S. manufacturing plants from 1977 to 2005 using Census plant-level data matched to detailed airline-route data from the U.S. Department of Transportation.
- **Core challenge.** Geographic distance between headquarters and plants is endogenous: firms choose where to locate plants and headquarters. A simple cross-sectional relation between distance and plant outcomes is therefore hard to interpret causally.
- **Main conceptual contribution.** The paper argues that the relevant object is not physical distance by itself but *travel time*, because travel time more directly captures how easy it is for headquarters to monitor plants and gather information.
- **Main empirical design.** The treatment is the introduction of a new airline route that reduces the travel time between a plant and its headquarters. The baseline specification is a plant-level difference-in-differences regression with plant and year fixed effects plus leave-one-out MSA-year and firm-year controls.
- **Main baseline result.** New airline routes that reduce travel time increase plant investment by about 8% to 9% and plant total factor productivity (TFP) by about 1.3% to 1.4%.
- **Economic magnitude.** The investment effect corresponds to an increase in capital expenditures of roughly \$158,000 to \$177,000 per treated plant, and the TFP effect corresponds to an increase in plant-level profits of about \$275,000 to \$296,000 (all in 1997 dollars).
- **Key identification result.** The estimates survive rich controls for local and firm shocks, dynamic pretrend checks, airline-merger and hub-opening subsamples, and same-last-leg / same-first-leg route changes.
- **Key heterogeneity result.** The treatment effect is stronger when headquarters is more time-constrained and in the earlier part of the sample, when alternative information technologies were weaker.
- **Extensive-margin result.** Time-constrained headquarters open plants closer to headquarters and close plants that are farther away; over time, firms have become more willing to open remote plants and close proximate ones.
- **Economic takeaway.** Transportation infrastructure affects more than shipping or commuting. It changes the internal allocation of capital and the efficiency of production by making monitoring and information transmission inside firms easier.

2 Data and measurement (key objects)

2.1 Why travel time and airline routes are useful

- The paper's starting point is that empirical work cares about proximity because proximity proxies for the ease of monitoring and acquiring information.

- Physical distance is an imperfect proxy for that object. A remote plant can be easy to visit if there is a fast direct flight, while a geographically closer plant can be hard to monitor if travel requires a long drive or inconvenient connections.
- Travel time is therefore a more direct proxy for the monitoring and information channel.
- The source of quasi-exogenous variation is the introduction of new airline routes that shorten the fastest feasible trip between headquarters and a plant.

Interpretation. The design does not say that airplanes themselves raise productivity. It says that reductions in the cost of *personal travel* make plant oversight easier.

2.2 Plant-level data and sample construction

The plant-level data come from three Census sources:

- the **Census of Manufactures (CMF)**, available every five years;
- the **Annual Survey of Manufactures (ASM)**, available in non-Census years;
- and the **Longitudinal Business Database (LBD)**, which provides annual establishment identifiers and firm affiliation.

Headquarters locations are constructed using the:

- **Auxiliary Establishment Survey (AES)**,
- **Standard Statistical Establishment List (SSEL)**,
- and payroll-based tie-breakers when multiple administrative offices exist.

Main sample construction choices:

- sample period: 1977–2005;
- require at least two consecutive years of plant data;
- exclude plants with imputed records, zero or missing employment, and firms that move headquarters during the sample period.

The raw selection leaves 1,332,824 plant-year observations. Using the paper’s main 10-year treatment window (five years before to five years after treatment) leaves a final sample of 1,291,280 plant-year observations.

Interpretation. The paper works with a very large plant-year panel and a treatment that is rare but far from trivial in scale.

2.3 Airline data and treatment definition

Airline data come from:

- the **T-100 Domestic Segment Database** for 1990–2005,
- and the **ER-586 Service Segment Data** for 1977–1989.

These data include every domestic flight segment, with information on:

- origin and destination airports,
- flight duration,
- and route structure.

Let the plant's ZIP code be P and headquarters' ZIP code be HQ . The fastest itinerary is the route that minimizes total travel time:

$$TT(P, HQ) = TT_{HQ \rightarrow \text{airport}}^{car} + TT^{flight} + TT_{\text{airport} \rightarrow P}^{car}.$$

The paper assumes:

- managers choose the fastest route and mode of transportation;
- one hour total is spent at origin and destination airports combined;
- each layover adds one hour.

A plant is **treated** when a newly introduced airline route lowers this minimum travel time.

Interpretation. Treatment is defined by an improvement in the best feasible trip, not merely by the existence of a new direct flight.

2.4 Outcome variables

Investment. The main investment measure is

$$Investment_{it} = \frac{\text{capital expenditures}_{it}}{\text{capital stock}_{it}},$$

with both numerator and denominator expressed in 1997 dollars. Investment is industry-adjusted by subtracting the median in the plant's three-digit SIC industry-year cell and winsorized at the 2.5th and 97.5th percentiles.

Productivity. The main productivity measure is TFP, estimated as the residual from a Cobb–Douglas production function run separately by industry and year:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \varepsilon_{it}.$$

Here, y is log output and k , l , and m are logs of capital, labor, and material inputs.

The paper also uses:

- **operating margin (OM)**,
- and **return on capital (ROC)**

as less structural productivity measures in robustness checks.

Interpretation. Investment captures the plant's capital-budget response; TFP captures whether easier access by headquarters improves production efficiency.

2.5 Key controls and covariates

The main time-varying controls are:

- **MSA-year controls:** the leave-one-out average of the dependent variable in the plant's MSA and year;
- **firm-year controls:** the leave-one-out average of the dependent variable across the firm's other plants in that year;
- plant **age:** $\ln(1 + \text{years in LBD})$;
- plant **size:** $\ln(\text{employees})$.

The paper sometimes also allows time shocks to differ by plant size through size $\times$ year fixed effects.

Interpretation. The leave-one-out MSA-year and firm-year controls are crucial: they purge shocks that are local to the plant's area or common to the firm as a whole.

2.6 Summary facts about the sample and treatments

Main descriptive facts:

- The sample contains 10,533 treated plants whose travel time falls because of new airline routes.
- Across all plants, the average distance to headquarters is 312 miles and average travel time is 126 minutes.
- For eventually treated plants, the average distance to headquarters is 854 miles and average travel time is 362 minutes.
- Across treated plants, the average one-way travel-time reduction is 103 minutes, or about 25%.

The paper classifies treatments into four categories:

- **indirect to indirect,**
- **indirect to direct,**
- **direct to direct,**
- **road to flight.**

The largest average travel-time reductions occur in the first two categories:

- indirect to indirect: 146 minutes,
- indirect to direct: 127 minutes.

Finally, the extensive-margin sample includes:

- 14,691 plant openings,
- and 22,281 plant closures.

Interpretation. The treatment is economically meaningful. The average improvement is not a few minutes; it is a large reduction in the cost of visiting remote plants.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

If firms operated with symmetric information and no agency problem, a reduction in travel time should not matter much for investment or productivity. The paper instead studies environments in which headquarters' proximity helps because:

- monitoring becomes easier,
- headquarters learns more about local projects,
- plant managers' incentives improve when oversight rises,
- and the transmission of soft information improves.

The paper also acknowledges possible alternatives that are close cousins of the same idea:

- headquarters may give better advice when it visits more often,
- plant morale may improve,
- or plants may receive more attention in budgeting simply because they become more salient.

Interpretation. The paper is about an internal-information channel broadly construed, not exclusively about a narrow textbook notion of monitoring.

3.2 Baseline difference-in-differences specification

The central regression is

$$y_{ijlt} = \alpha_i + \alpha_t + \beta Treatment_{it} + \delta' X_{ijlt} + \varepsilon_{ijlt}, \quad (1)$$

where:

- i indexes plants,
- j indexes firms,
- l indexes plant location,
- t indexes years,
- y_{ijlt} is plant investment or productivity,
- α_i are plant fixed effects,
- α_t are year fixed effects,
- $Treatment_{it}$ equals 1 once a new airline route has reduced travel time to headquarters,
- and X_{ijlt} contains MSA-year controls, firm-year controls, age, and size.

Interpretation. The coefficient β compares changes in treated plants after the route introduction with changes in plants that are not yet treated, while holding constant plant-specific levels and year-specific common shocks.

3.3 Productivity equation

TFP is constructed from:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \varepsilon_{it}, \quad (2)$$

estimated separately for each industry-year cell.

Interpretation.

- The residual ε_{it} is the plant's productivity relative to others in the same industry-year.
- The paper estimates TFP by OLS in the main specification and checks robustness using the Olley–Pakes and Levinsohn–Petrin procedures.

3.4 Dynamic event-time specification

To test pretrends and adjustment timing, the paper replaces the treatment dummy with event-time bins measured in months relative to the route introduction:

$$y_{it} = \alpha_i + \alpha_t + \sum_s \beta_s D_{it}^{(s)} + \delta' X_{it} + \varepsilon_{it}, \quad (3)$$

where the $D_{it}^{(s)}$ indicators correspond to intervals such as:

- $(-12, -6)$ months,
- $(-6, 0)$ months,
- $(0, 6)$ months,
- $(6, 12)$ months,
- $(12, 18)$ months,
- $(18, 24)$ months,
- $(24, 30)$ months,
- and $(30+)$ months.

Interpretation. This is an event-study style design. If airline routes are introduced because of a preexisting plant shock, the coefficients before treatment should already be positive.

3.5 Heterogeneity specifications

The paper studies two main forms of heterogeneity.

Headquarters' time constraints. The treatment is interacted with indicators for whether headquarters is more or less time-constrained, using:

- managers / plants,
- and managers / total distance to all plants.

Information technology over time. The treatment is interacted with time-period indicators:

- pre-1986,
- 1986–1995,
- post-1995.

Interpretation. If easier travel matters because it helps managers allocate scarce attention and gather soft information, the effect should be strongest when time is scarce and when remote communication technology is weak.

3.6 Robustness specifications tied to identification

The paper estimates several alternative versions of the treatment:

- only routes introduced in airline **hub openings**,
- only routes introduced after airline **mergers**,
- only indirect flights with the **same last leg**,
- only indirect flights with the **same first leg**.

Interpretation. These variants isolate route changes that are especially unlikely to be caused by a shock to one plant or to one plant-headquarters pair.

3.7 Extensive-margin location regressions

To study openings and closures, the paper regresses the logarithm of plant-headquarters distance or travel time on time-constraint measures and period indicators:

$$\ln Distance_{it} = \alpha + \beta HQConstraint_{jt-1} + \gamma' Controls_{it} + \lambda_t + \varepsilon_{it}. \quad (4)$$

It also estimates versions where the key regressors are period dummies for 1986–1995 and post-1995.

Interpretation. These regressions ask whether firms choose plant locations differently when monitoring is harder or when communication technology changes over time.

3.8 Estimation, treatment windows, and standard errors

Main estimation choices:

- plant fixed effects and year fixed effects in the baseline;
- standard errors clustered at the MSA level;
- 10-year treatment window around the treatment date in the main sample.

The paper shows that results are similar when:

- clustering at the firm level,
- clustering two ways,
- collapsing to before-versus-after periods,
- or changing the treatment window length.

Interpretation. The results do not rely on a particular treatment window or one fragile inference choice.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The paper does *not* use a formal instrument. Identification comes from:

- within-plant changes over time,
- the timing of new airline routes,
- and controls that absorb local and firm-level shocks.

The identifying comparison is: when a plant becomes easier to reach because of a new route, do investment and productivity rise relative to otherwise similar untreated plants?

Interpretation. This is a carefully structured difference-in-differences design rather than an IV paper.

4.2 Why simple reverse causality is weak

Could plant investment itself cause airline routes to open? The paper argues that the simple reverse-causality story is weak because:

- airlines respond to route profitability at the airport-pair level, not to plant-level investment alone;
- treatment depends on travel-time improvements across airport networks;
- and dynamic regressions show no pre-existing increase in investment before treatment.

Interpretation. If routes were opened because plants were already booming, the event-study coefficients before treatment should already move.

4.3 Local shocks

A major concern is that local shocks in the plant's area might both:

- increase plant investment,

- and attract new airline service.

The paper addresses this in two ways:

- **MSA-year controls** absorb average shocks in the plant's location;
- a robustness sample keeps only route changes where the **last leg** of the trip remains unchanged.

Interpretation. If a Memphis-area boom explains the result, other Memphis plants with different headquarters should look similar. The leave-one-out MSA-year control is designed exactly for that.

4.4 Firm-level shocks

A second concern is that some firm-wide shock might raise investment across all plants and also coincide with route introductions.

The paper deals with this by:

- adding **firm-year controls**,
- and studying route changes where the **first leg** of the trip remains unchanged.

Interpretation. If a Boston firm's overall growth is the true driver, then untreated Boston-linked plants in the same firm should move too, and the firm-year control will absorb that.

4.5 Plant-specific or pair-specific shocks

The hardest concern is a shock that is specific to one plant or one plant-headquarters pair. Because such a shock is collinear with treatment, MSA-year and firm-year controls cannot fully remove it.

The paper addresses this using three ideas:

- **dynamic pretrend tests:** no evidence of a treatment effect before the route opens;
- **hub openings and airline mergers:** it is implausible that one plant causes an airline merger or a hub opening;
- **same-last-leg / same-first-leg routes:** one plant is unlikely to engineer a flight elsewhere in the network that leaves its own airport link unchanged.

Interpretation. This is the paper's tightest identification argument.

4.6 Additional robustness checks

The paper also shows that the main conclusions survive:

- restricting the sample to multiunit firms,
- restricting the sample to eventually treated plants,
- adding a second treatment for *increases* in travel time when routes are terminated,
- using alternative treatment windows,

- and using alternative productivity measures and estimators.

Interpretation. The effect is not an artifact of single-unit firms, control-group composition, or one specific productivity definition.

4.7 What the paper can and cannot distinguish

A central limitation is that the paper identifies variation in *travel time*, not direct variation in monitoring effort or information flows themselves.

So the paper can strongly establish:

- reduced travel time raises investment and productivity,
- and the pattern is consistent with proximity helping headquarters.

But it cannot perfectly separate whether the mechanism is:

- monitoring,
- better advice,
- improved morale,
- more salience in capital budgeting,
- or some other soft-information channel.

Interpretation. The paper is strongest on the existence and size of the proximity effect; it is more agnostic about the precise organizational micro-mechanism inside that effect.

4.8 Relation to the literature

The paper connects several literatures:

- geographic proximity and information in finance,
- monitoring inside organizations,
- internal capital markets,
- and plant-level productivity.

Its distinctive contribution is to move from arm's-length transactions to *within-firm* geography and to do so using plant-level treatment variation.

5 Tables and figures

5.1 Table I and Table II: sample characteristics and treatment magnitudes

Read:

- Table I shows that treated plants are much farther from headquarters than average plants: 854 miles versus 312 miles.
- Average travel time for treated plants is 362 minutes before treatment, compared with 126 minutes in the full sample.
- Table II shows that the average one-way reduction in travel time is 103 minutes (25%).
- The largest reductions come from indirect-to-indirect and indirect-to-direct route changes.

Economic message. The treatment is large enough to plausibly change real monitoring behavior. It is not a tiny marginal convenience shock.

TABLE I
SUMMARY STATISTICS: PLANTS

	(1) All plants	(2) Eventually new airline route	(3) No new airline route
Total value of shipments	50,196 (360,930)	75,752 (222,685)	48,721 (367,270)
Capital stock	20,710 (106,473)	33,615 (118,024)	19,965 (105,719)
Employees	213 (568)	300 (638)	208 (564)
Distance to headquarters (miles)	312 (563)	854 (616)	281 (544)
Travel time (minutes)	126 (170)	362 (135)	113 (162)
Number of observations	1,291,280	70,467	1,220,813

Notes. "All plants" refers to all plants in the sample. "Eventually new airline route" refers to plants that are treated during the sample period, that is, plants whose travel time to headquarters is reduced through the introduction of a new airline route. "No new airline route" refers to plants that are not treated during the sample period. Total value of shipments and capital stock are expressed in 1997 dollars (in 1,000's) using four-digit SIC deflators from the NBER-CES Manufacturing Industry Database. Capital stock is constructed using the perpetual inventory method described in Appendix B. Employees is the number of employees of the plant. Distance to headquarters is the great-circle distance between the plant's ZIP code and the ZIP code of headquarters (in miles). Travel time is the total travel time based on the fastest route and means of transportation (car or plane) between the plant's ZIP code and the ZIP code of headquarters (in minutes). All figures are sample means based on unadjusted (i.e., nonwinsorized) distributions. Standard deviations are in parentheses. The sample period is from 1977 to 2005.

TABLE II
SUMMARY STATISTICS: TRAVEL TIME REDUCTIONS

	(1) All	(2) Indirect to indirect	(3) Indirect to direct	(4) Direct to direct	(5) Road to flight
Distance to headquarters (miles)	854	1,211	942	726	191
Travel time before (minutes)	417	566	466	338	253
Travel time after (minutes)	314	420	339	266	206
Δ travel time (minutes)	-103	-146	-127	-72	-47
Δ travel time (%)	-25	-26	-27	-21	-19
Number of observations	10,533	1,911	3,469	4,544	609

Notes. "All" refers to all treated plants, that is, plants whose travel time to headquarters is reduced through the introduction of a new airline route during the sample period. "Indirect to indirect" refers to the subset of treated plants for which a new indirect flight using a different route replaces a previously optimal indirect flight. "Indirect to direct" refers to the subset of treated plants for which a new direct flight replaces a previously optimal indirect flight. "Direct to direct" refers to the subset of treated plants for which a new direct flight using a different route replaces a previously optimal direct flight. "Road to flight" refers to the subset of treated plants for which a new direct or indirect flight replaces car travel as the previously optimal means of transportation. Distance to headquarters is the great-circle distance between the plant's ZIP code and the ZIP code of headquarters (in miles). Travel time before is the total travel time between the plant's ZIP code and the ZIP code of headquarters based on the fastest route and means of transportation (car or plane) before the introduction of the new airline route (in minutes). Travel time after is defined accordingly. Δ travel time is the difference between travel time after and travel time before, expressed either in minutes or as a percentage of travel time before. All figures are sample means across all plants. The sample period is from 1977 to 2005.

5.2 Table III: baseline effect of new airline routes

Read:

- The treatment coefficient is 0.008 in the simplest investment regression and 0.009 once MSA-year and firm-year controls are added.
- The corresponding TFP coefficient is 0.014 in the simplest productivity regression and 0.013 once controls are added.
- The baseline investment effect is therefore about 8% to 9% relative to the sample mean of investment.
- The productivity effect is about 1.3% to 1.4%.

Economic message. This is the core causal result: improving headquarters’ access to a plant raises both capital spending and efficiency.

TABLE III
THE EFFECT OF NEW AIRLINE ROUTES ON PLANT INVESTMENT AND PRODUCTIVITY

Dependent variable	Investment			TFP		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.008*** (0.001)	0.009*** (0.001)	0.010*** (0.001)	0.014*** (0.003)	0.013*** (0.003)	0.020*** (0.003)
MSA-year		0.153*** (0.022)	0.148*** (0.022)		0.080*** (0.012)	0.080*** (0.012)
Firm-year		0.205*** (0.006)	0.205*** (0.006)		0.186*** (0.005)	0.186*** (0.005)
Age		−0.060*** (0.002)	−0.061*** (0.002)		0.015*** (0.002)	0.015*** (0.002)
Size		0.029*** (0.001)			0.012*** (0.002)	
Plant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Size × year fixed effects	No	No	Yes	No	No	Yes
R-squared	0.39	0.41	0.41	0.60	0.61	0.61
Number of observations	1,291,280	1,291,280	1,291,280	1,291,280	1,291,280	1,291,280

Notes. Investment is the ratio of capital expenditures to capital stock, which is industry-adjusted by subtracting the industry median across all plants in a given three-digit industry and year. Total factor productivity (TFP) is the residual from estimating a log-linear Cobb-Douglas production function by ordinary least squares for each three-digit industry and year at the plant level. Investment and TFP are winsorized at the 2.5th and 97.5th percentiles of their empirical distribution. Treatment is a dummy variable equals 1 if a new airline route that reduces the travel time between a plant and its headquarters has been introduced. MSA-year and firm-year indicate the mean of the dependent variable in the plant’s MSA and firm, respectively, excluding the plant itself. Age is the natural logarithm of one plus the number of years since the plant has been in the LB is the natural logarithm of the number of employees of the plant. Standard errors are clustered at the MSA level. The sample period is from 1977 to 2005. Standard errors in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

5.3 Table IV: dynamic effects and pretrends

Read:

- There is no statistically meaningful effect on investment or TFP in the months before treatment.
- Investment becomes significant after 6–12 months and reaches about 0.013–0.014 from 12 to 30 months after treatment.
- TFP rises with a longer lag and reaches about 0.020 in the 18–30 month window.

Economic message. The absence of pretrends strengthens identification, and the lagged response is economically intuitive: investment reacts first and productivity follows.

TABLE IV
DYNAMIC EFFECTS OF NEW AIRLINE ROUTES

dependent variable	(1) Investment	(2) TFP
treatment (−12 m, −6 m)	−0.000 (0.003)	−0.001 (0.005)
treatment (−6 m, 0 m)	−0.001 (0.002)	−0.001 (0.004)
treatment (0 m, 6 m)	0.003 (0.003)	0.001 (0.005)
treatment (6 m, 12 m)	0.005** (0.002)	0.006 (0.005)
treatment (12 m, 18 m)	0.013*** (0.003)	0.012** (0.005)
treatment (18 m, 24 m)	0.014*** (0.002)	0.020*** (0.004)
treatment (24 m, 30 m)	0.014*** (0.003)	0.020*** (0.005)
treatment (30 m +)	0.009*** (0.002)	0.013*** (0.004)
SA-year	0.153*** (0.022)	0.080*** (0.012)
rm-year	0.205*** (0.006)	0.186*** (0.005)
ge	−0.060*** (0.002)	0.015*** (0.002)
ze	0.029*** (0.001)	0.012*** (0.002)
plant fixed effects	Yes	Yes
year fixed effects	Yes	Yes
R-squared	0.41	0.61
number of observations	1,291,280	1,291,280

Notes. Treatment (−12 m, −6 m) is a dummy variable that equals 1 if the plant-year observation is recorded between 6 and 12 months before the introduction of the new

5.4 Figure I and Tables V–VI: robustness to plant-specific-shock stories

Read:

- Figure I shows a large spike in the number of treated plants in the exact year when an airline opens a hub or completes a merger.
- Table V shows that hub treatments and merger treatments both generate economically large effects. For investment, the coefficients are about 0.017 for hub treatments and 0.014 for merger treatments.
- Table VI shows that the treatment survives when the same last leg or same first leg of the route remains unchanged. The investment coefficients are about 0.012 and 0.013, respectively.

Economic message. The main findings are not driven by plant-specific lobbying or by a single

plant causing a route change tailored to itself.

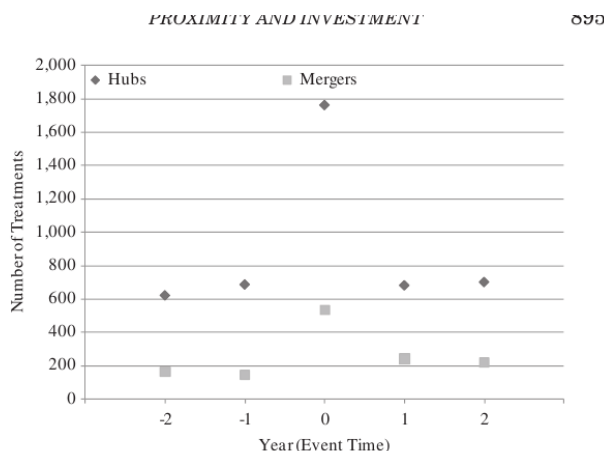


FIGURE I
Number of New Treatments around Hub Openings and Airline Mergers

TABLE V
HUB OPENINGS AND AIRLINE MERGERS

Dependent variable	(1) Investment	(2) TFP
Treatment (hub)	0.017*** (0.002)	0.019*** (0.005)
Treatment (merger)	0.014** (0.006)	0.018** (0.009)
Treatment (other)	0.008*** (0.001)	0.011*** (0.003)
SA-year	0.153*** (0.022)	0.080*** (0.012)
Firm-year	0.205*** (0.006)	0.186*** (0.005)
Age	-0.060*** (0.002)	0.015*** (0.002)
Size	0.029*** (0.001)	0.012*** (0.002)
Plant fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
R-squared	0.41	0.61
Number of observations	1,291,280	1,291,280

Notes. Treatment (hub) and treatment (merger) are dummy variables that equal 1 if the treatment dummy equals 1 and the new airline route is introduced by an airline in the same year as it opens a new hub or merges its operations with another airline, respectively. Treatment (other) is a dummy variable that equals 1 if the treatment dummy equals 1 and the treatment is not a hub or merger treatment as

TABLE VI
NEW AIRLINE ROUTES WITH SAME LAST LEG OR SAME FIRST LEG

Dependent variable	(1) Investment	(2) TFP
Treatment (same last leg)	0.012*** (0.004)	0.014* (0.008)
Treatment (same first leg)	0.013*** (0.004)	0.015** (0.008)
Treatment (other)	0.009*** (0.001)	0.012*** (0.003)
MSA-year	0.153*** (0.022)	0.080*** (0.012)
Firm-year	0.205*** (0.006)	0.186*** (0.005)
Age	-0.060*** (0.002)	0.015*** (0.002)
Size	0.029*** (0.001)	0.012*** (0.002)
Plant fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
R-squared	0.41	0.61
Number of observations	1,291,280	1,291,280

Notes. Treatment (same last leg) and treatment (same first leg) are dummy variables that equal 1 if the treatment dummy equals 1 and the new airline route operates the same last leg or the same first leg, respectively, as the previously optimal airline route. Treatment (other) is a dummy variable that equals 1 if the treatment dummy equals 1 and the new airline route operates neither the same last leg nor the

5.5 Table VII: alternative control groups

Read:

- Restricting the sample to multiunit firms leaves the treatment effect essentially unchanged.
- Restricting the sample to eventually treated plants also leaves the result intact.
- Adding an “increase in travel time” dummy for terminated routes yields a negative coefficient on travel-time increases but does not alter the main positive treatment effect.

Economic message. The results do not depend on including single-unit firms, never-treated plants, or ignoring route terminations.

TABLE VII
ALTERNATIVE CONTROL GROUPS

Dependent variable	Multiunit firms		Eventually treated plants		Increase in travel time	
	(1) Investment	(2) TFP	(3) Investment	(4) TFP	(5) Investment	(6) TFP
Treatment	0.009*** (0.001)	0.012*** (0.003)	0.011*** (0.002)	0.010*** (0.003)	0.009*** (0.001)	0.010*** (0.003)
Increase in travel time					−0.005** (0.002)	−0.006** (0.002)
MSA-year	0.133*** (0.019)	0.090*** (0.015)	0.084** (0.042)	0.084** (0.041)	0.153*** (0.022)	0.084** (0.01)
Firm-year	0.207*** (0.006)	0.186*** (0.005)	0.257*** (0.016)	0.293*** (0.016)	0.205*** (0.006)	0.186*** (0.006)
Age	−0.072*** (0.002)	0.013*** (0.003)	−0.047*** (0.004)	0.038*** (0.009)	−0.060*** (0.002)	0.013*** (0.003)
Size	0.026*** (0.001)	0.020*** (0.002)	0.027*** (0.002)	0.025*** (0.006)	0.029*** (0.001)	0.020*** (0.002)
Plant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.37	0.61	0.32	0.65	0.41	0.61
Number of observations	825,097	825,097	70,467	70,467	1,282,228	1,282,228

Notes. In columns (1) and (2), the sample is restricted to plants that belong to multiunit firms consisting of more than one establishment. In columns (3) and (4), the sample is restricted to plants that are eventually treated—that is, plants whose travel time to headquarters is reduced through the introduction of a new airline route—during the sample period. Increase in travel time is a dummy variable that equals 1 if the travel time to headquarters increases during the sample period due to the termination of an existing route. All other variables are defined in Table III. Standard errors are clustered at the MSA level. The sample period is from 1977 to 2005. Standard errors are in parentheses.

5.6 Tables VIII and IX: why the effect is stronger in some cases

Read:

- Table VIII shows that the effect is much larger when headquarters is more time-constrained. For example, investment rises by about 0.012–0.013 for high-constraint headquarters and only 0.005–0.006 for low-constraint headquarters.
- Table IX shows that the treatment effect declines over time. For investment, the coefficient falls from about 0.013 before 1986 to about 0.005 after 1995.

Economic message. These heterogeneity patterns are exactly what we would expect if personal travel matters because it helps time-constrained managers gather information in an era before strong digital communication tools.

TABLE VIII
HEADQUARTERS' TIME CONSTRAINTS

Dependent variable	Managers/plants		Managers/total distance	
	(1) Investment	(2) TFP	(3) Investment	(4) TFP
Treatment $\times$ high time constraints	0.012*** (0.002)	0.015*** (0.004)	0.013*** (0.002)	0.015*** (0.003)
Treatment $\times$ low time constraints	0.006*** (0.002)	0.010** (0.004)	0.005** (0.002)	0.009* (0.005)
MSA-year	0.153*** (0.022)	0.080*** (0.012)	0.153*** (0.022)	0.080*** (0.012)
Firm-year	0.205*** (0.006)	0.186*** (0.005)	0.205*** (0.006)	0.186*** (0.005)
Age	-0.060*** (0.002)	0.015*** (0.002)	-0.060*** (0.002)	0.015*** (0.002)
Size	0.029*** (0.001)	0.012*** (0.002)	0.029*** (0.001)	0.012*** (0.002)
Plant fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
R-squared	0.41	0.61	0.41	0.61
Number of observations	1,291,280	1,291,280	1,291,280	1,291,280

Notes. In columns (1) and (2), headquarters' time constraints are measured as the number of (white-collar) employees at headquarters divided by the total number of plants of the company ("managers/plants"). In columns (3) and (4), headquarters' time constraints are measured as the number of (white-collar) employees at headquarters divided by the total distance (in miles) between headquarters and all of the company's plants ("managers/total distance"). High time constraints is a dummy variable that equals 1 if the measure of headquarters' time constraint lies above the median value across all treated plants in the year prior to the treatment. Low time constraints is defined analogously. All other

TABLE IX
INNOVATIONS IN INFORMATION TECHNOLOGY

Dependent variable	(1) Investment	(2) TFP
Treatment $\times$ pre 1986	0.013*** (0.002)	0.019*** (0.004)
Treatment $\times$ between 1986 and 1995	0.010*** (0.002)	0.012*** (0.004)
Treatment $\times$ post 1995	0.005** (0.002)	0.009* (0.005)
MSA-year	0.153*** (0.022)	0.080*** (0.012)
Firm-year	0.205*** (0.006)	0.186*** (0.005)
Age	-0.060*** (0.002)	0.015*** (0.002)
Size	0.029*** (0.001)	0.012*** (0.002)
Plant fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
R-squared	0.41	0.61
Number of observations	1,291,280	1,291,280

Notes. Pre 1986, between 1986 and 1995, and post 1995 are dummy variables that equal 1 if the plant-year observation lies within the specified time interval. All other variables are defined in Table III. Standard errors are clustered at the MSA level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

5.7 Tables X and XI: extensive-margin evidence on plant location

Read:

- Table X shows that firms with more time-constrained headquarters open plants closer to headquarters and close plants that are farther away.
- Table XI shows that over time the distance between headquarters and newly opened plants rises, while the plants closed by firms become relatively more proximate.
- In Table XI, the coefficients for plant openings are about 0.185–0.190 in 1986–1995 and about 0.297–0.305 after 1995, relative to the pre-1986 period.

Economic message. The same friction that affects intensive-margin investment also appears to shape firms' location choices on the extensive margin.

TABLE X
HEADQUARTERS' TIME CONSTRAINTS AND PLANT LOCATION

Sample Dependent variable	Plant openings				Plant closures			
	Geographical distance		Travel time		Geographical distance		Travel time	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Managers/plants	0.148*** (0.027)		0.121*** (0.022)		-0.209*** (0.019)		-0.192*** (0.015)	
Managers/total distance		0.424*** (0.010)		0.313*** (0.009)		-0.428*** (0.009)		-0.313*** (0.009)
Firm size	0.557*** (0.014)	0.314*** (0.014)	0.439*** (0.011)	0.259*** (0.011)	0.647*** (0.014)	0.362*** (0.015)	0.543*** (0.010)	0.313*** (0.010)
Firm age	0.523*** (0.041)	0.384*** (0.036)	0.544*** (0.037)	0.442*** (0.033)	0.728*** (0.100)	0.496*** (0.089)	0.701*** (0.089)	0.523*** (0.089)
Size	-0.279*** (0.018)	-0.173*** (0.016)	-0.246*** (0.014)	-0.169*** (0.013)	-0.282*** (0.013)	-0.179*** (0.012)	-0.251*** (0.010)	-0.169*** (0.010)
Age					-0.304*** (0.034)	-0.257*** (0.031)	-0.284*** (0.025)	-0.257*** (0.025)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.27	0.39	0.28	0.38	0.38	0.48	0.40	0.41
Number of observations	14,691	14,691	14,691	14,691	22,281	22,281	22,281	22,281

Notes. The sample of plant openings (closures) in columns (1)–(4) ((5)–(8)) includes all sample plants that are coded as “birth” (“death”) in the LBD. Geographical distance is the natural logarithm of one plus the great-circle distance between the plant’s ZIP code and the ZIP code of headquarters. Travel time is the natural logarithm of one plus the travel time based on the fastest route and means of transportation (car or plane) between the plant’s ZIP code and the ZIP code of headquarters. Firm size is the natural log of the total number of employees of the parent company to which the plant belongs. Firm age is the natural logarithm of one plus the number of years the parent company has been in the LBD. All other variables are defined in Table III. Standard errors are clustered at the MSA level. The sample period is from 1977 to 2005. Standard errors are in parentheses.

TABLE XI
INNOVATIONS IN INFORMATION TECHNOLOGY AND PLANT LOCATION

Sample Dependent variable	Plant openings		Plant closures	
	(1) Geographical distance	(2) Travel time	(3) Geographical distance	(4) Travel time
Between 1986 and 1995	0.185*** (0.063)	0.190*** (0.049)	-0.126** (0.057)	-0.104** (0.047)
Post 1995	0.297*** (0.078)	0.305*** (0.063)	-0.197** (0.077)	-0.168*** (0.064)
Firm size	0.580*** (0.014)	0.461*** (0.011)	0.686*** (0.014)	0.580*** (0.010)
Firm age	0.451*** (0.035)	0.459*** (0.029)	0.338*** (0.061)	0.341*** (0.051)
Size	-0.319*** (0.017)	-0.283*** (0.014)	-0.317*** (0.013)	-0.282*** (0.010)
Age			-0.314*** (0.033)	-0.294*** (0.025)
R-squared	0.26	0.27	0.37	0.39
Number of observations	14,691	14,691	22,281	22,281

Notes. Between 1986 and 1995 and post 1995 are dummy variables that equal 1 if the plant is opened (columns (1)–(2)) or closed (columns (3)–(4)) within the specified time interval. All other variables are defined in Tables III and X. Standard errors are clustered at the MSA level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

6 Short summary

- The paper asks whether easier access from headquarters to plants changes real investment and productivity inside firms.
- It uses new airline routes as a source of plausibly exogenous variation in travel time between headquarters and plants.
- Treated plants increase investment by about 8% to 9% and TFP by about 1.3% to 1.4%.
- Event-study results show no pretrends and a gradual post-treatment rise in investment and productivity.
- The estimates survive MSA-year and firm-year controls, hub-opening and merger treatments, same-last-leg and same-first-leg samples, and alternative control groups.
- The effect is strongest when headquarters is more time-constrained and in the earlier years of the sample.
- Firms also appear to choose plant locations in ways consistent with monitoring frictions: constrained headquarters open plants nearby and close distant ones.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that proximity to headquarters matters for plant-level investment and productivity. When the introduction of a new airline route shortens travel time between headquarters and a plant, the plant invests more and becomes more productive.

7.2 Economic intuition

The basic intuition is simple. Large firms do not allocate capital to plants in a frictionless way. Headquarters needs time and attention to monitor plants, collect soft information, advise managers, and understand where investment opportunities are best. When travel becomes easier, those frictions fall, and the internal allocation of capital improves.

7.3 Takeaway

The paper's strongest message is that transport infrastructure changes firm boundaries from the inside. It affects not only where firms can ship goods, but also how well headquarters can oversee distant operations. That makes this paper an important contribution to corporate finance, productivity, and organizational economics.